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Namespace Uralstech.UAI.LiteRTLM

Classes

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[TokenUnion](#)

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Structs

[BenchmarkInfo.Turn](#)

Benchmarks for a prefill or decode turn.

Enums

[ActivationDataType](#)

[ConstraintProviderType](#)

[ConstraintType](#)

[InputDataType](#)

Represents the type of input data.

[LogSeverity](#)

Represents the log severity / level.

[SamplerType](#)

Represents the type of sampler.

[TokenUnionType](#)

Represents the type of a TokenUnion.

Delegates

[StreamCallback](#)

Callback for streaming responses.

Enum ActivationDataType

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public enum ActivationDataType
```

Fields

Float16 = 1

Use float16 as the activation data type.

Float32 = 0

Use float32 as the activation data type.

Int16 = 2

Use int16 as the activation data type.

Int8 = 3

Use int8 as the activation data type.

Class BackendNames

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public static class BackendNames
```

Inheritance

object ← BackendNames

Fields

CPU

```
public const string CPU = "cpu"
```

Field Value

string

CPUArtisan

```
public const string CPUArtisan = "cpu_artisan"
```

Field Value

string

GPU

```
public const string GPU = "gpu"
```

Field Value

string

GPUArtisan

```
public const string GPUArtisan = "gpu_artisan"
```

Field Value

string

GoogleTensorArtisan

```
public const string GoogleTensorArtisan = "google_tensor_artisan"
```

Field Value

string

NPU

```
public const string NPU = "npu"
```

Field Value

string

Class BenchmarkInfo

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class BenchmarkInfo : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← BenchmarkInfo

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Methods

GetDecodeTurns()

Returns the benchmark information for each decode turn.

```
public BenchmarkInfo.Turn[] GetDecodeTurns()
```

Returns

[Turn\[\]](#)

GetPrefillTurns()

Returns the benchmark information for each prefill turn.

```
public BenchmarkInfo.Turn[] GetPrefillTurns()
```

Returns

[Turn\[\]](#)

GetTimeToFirstToken()

Returns the time to the first token in seconds.

```
public double GetTimeToFirstToken()
```

Returns

double

The time to the first token in seconds.

Remarks

Note that the time to first token does not include initialization time. It is the sum of the prefill time for the first turn and the time spent decoding the first token.

GetTotalInitTime()

Returns the total initialization time in seconds.

```
public double GetTotalInitTime()
```

Returns

double

The total initialization time in seconds.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Struct BenchmarkInfo.Turn

Namespace: [Uralstech.UAI.LiteRTLM](#)

Benchmarks for a prefill or decode turn.

```
public readonly struct BenchmarkInfo.Turn
```

Constructors

Turn(int, double)

```
public Turn(int tokenCount, double tokensPerSecond)
```

Parameters

tokenCount int

tokensPerSecond double

Fields

TokenCount

The prefill/decode token count for this turn.

```
public readonly int TokenCount
```

Field Value

int

TokensPerSecond

The prefill/decode tokens per second for this turn.

```
public readonly double TokensPerSecond
```

Field Value

double

Enum ConstraintProviderType

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public enum ConstraintProviderType
```

Fields

LlGuidance = 1

Enum ConstraintType

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public enum ConstraintType
```

Fields

JsonSchema = 2

None = 0

Regex = 1

Class Conversation

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Conversation : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← Conversation

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

Conversation(Engine, ConversationConfig?)

Creates a managed wrapper around a LiteRT LM conversation. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public Conversation(Engine engine, ConversationConfig? config = null)
```

Parameters

engine [Engine](#)

The engine to create the conversation from.

config [ConversationConfig](#)

The conversation configuration to use. If [null](#), the default configuration is used.

Exceptions

[InvalidOperationException](#)

Thrown if the native object could not be created.

Methods

CancelProcess()

Cancels the ongoing inference process for asynchronous inference.

```
public void CancelProcess()
```

Clone()

Clones a LiteRT LM conversation, duplicating its prefilled state. The caller is responsible for disposing the cloned wrapper using [Dispose\(\)](#).

```
public Conversation? Clone()
```

Returns

[Conversation](#)

The cloned conversation wrapper, or [null](#) on failure.

GetBenchmarkInfo()

Retrieves benchmark information from the conversation. The caller is responsible for disposing the returned wrapper.

```
public BenchmarkInfo? GetBenchmarkInfo()
```

Returns

[BenchmarkInfo](#)

The benchmark information wrapper, or [null](#) on failure.

GetTokenCount()

Gets the number of tokens in the conversation KV cache (prefill + decode).

```
public int GetTokenCount()
```

Returns

int

The number of tokens, or a negative value on failure.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

RenderMessageToString(string)

Renders the message into a string according to the template.

```
public string? RenderMessageToString(string messageJson)
```

Parameters

messageJson string

A JSON string representing the message to render.

Returns

string

The rendered string, or [null](#) on failure.

Remarks

This function does not need to be called for actual message sending, as [Send Message\(string, string?, ConversationOptionalArgs?\)](#) and [SendMessageStream\(Stream Callback, string, string?, ConversationOptionalArgs?\)](#) render messages internally.

RenderPrefaceToString()

Renders the preface into a string according to the template.

```
public string? RenderPrefaceToString()
```

Returns

string

The rendered string, or [null](#) on failure.

SendMessage(string, string?, ConversationOptionalArgs?)

Sends a message to the conversation and returns the response. This is a blocking call.

```
public JsonResponse? SendMessage(string messageJson, string? extraContext = null,
    ConversationOptionalArgs? optionalArgs = null)
```

Parameters

messageJson string

A JSON string representing the message to send.

extraContext string

A JSON string representing the extra context to use.

optionalArgs [ConversationOptionalArgs](#)

The optional arguments to use.

Returns

[JsonResponse](#)

A JSON response wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

SendMessageStream(StreamCallback, string, string?, ConversationOptionalArgs?)

Sends a message to the conversation and streams the response through a callback. This is a non-blocking call that invokes the callback from a background thread for each chunk.

```
public int SendMessageStream(StreamCallback callback, string messageJson, string? extraContext = null, ConversationOptionalArgs? optionalArgs = null)
```

Parameters

callback [StreamCallback](#)

The callback function ([StreamCallback](#)) that receives response chunks.

messageJson string

A JSON string representing the message to send.

extraContext string

A JSON string representing the extra context to use.

optionalArgs [ConversationOptionalArgs](#)

The optional arguments to use.

Returns

int

0 on success, non-zero on failure to start the stream.

Class ConversationConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class ConversationConfig : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← ConversationConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

ConversationConfig()

Creates a managed wrapper around a LiteRT LM conversation configuration. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public ConversationConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetConstraintProvider(ConstraintProviderType?)

Sets the constraint provider type for this conversation config.

```
public void SetConstraintProvider(ConstraintProviderType? providerType)
```

Parameters

providerType [ConstraintProviderType?](#)

The constraint provider type to use, or [null](#) to unset.

SetEnableConstrainedDecoding(bool)

Sets whether to enable constrained decoding for this conversation configuration.

```
public void SetEnableConstrainedDecoding(bool enableConstrainedDecoding)
```

Parameters

enableConstrainedDecoding bool

Whether to enable constrained decoding.

SetExtraContext(string)

Sets the extra context for the conversation preface.

```
public void SetExtraContext(string extraContextJson)
```

Parameters

extraContextJson string

A JSON string representing the extra context object.

SetFilterChannelContentFromKvCache(bool)

Sets whether to filter channel content from the KV cache.

```
public void SetFilterChannelContentFromKvCache(bool filterChannelContentFromKvCache)
```

Parameters

filterChannelContentFromKvCache bool

Whether to filter channel content.

SetMessages(string)

Sets the initial messages for this conversation configuration.

```
public void SetMessages(string messagesJson)
```

Parameters

messagesJson string

The initial messages in JSON array format.

SetPromptTemplate(string)

Sets the prompt template for this conversation config.

```
public void SetPromptTemplate(string promptTemplate)
```

Parameters

promptTemplate string

The prompt template string (e.g. Jinja template). If not set, use the default provided by the model or the engine.

SetSessionConfig(SessionConfig)

Sets the session config for this conversation configuration.

```
public void SetSessionConfig(SessionConfig sessionConfig)
```

Parameters

sessionConfig [SessionConfig](#)

The session config to use.

SetStreamToolCalls(bool, string)

Sets whether to stream tool call tokens.

```
public void SetStreamToolCalls(bool streamToolCalls, string channelName)
```

Parameters

streamToolCalls bool

Whether to stream tool call tokens.

channelName string

The channel name to use for tool call tokens.

SetSystemMessage(string)

Sets the system message for this conversation configuration.

```
public void SetSystemMessage(string systemMessageJson)
```

Parameters

systemMessageJson string

The system message in JSON format.

SetThinkingConfig(ThinkingConfig?)

Sets the thinking config for this conversation config.

```
public void SetThinkingConfig(ThinkingConfig? thinkingConfig)
```

Parameters

thinkingConfig [ThinkingConfig](#)

The thinking config to set. If [null](#), clears any previously set thinking config.

SetTools(string)

Sets the tools for this conversation configuration.

```
public void SetTools(string toolsJson)
```

Parameters

toolsJson string

The tools description in JSON array format.

Class ConversationOptionalArgs

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class ConversationOptionalArgs : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← ConversationOptionalArgs

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

ConversationOptionalArgs()

Creates a managed wrapper around the optional arguments for conversation APIs. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public ConversationOptionalArgs()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetConstraint(ConstraintType, string)

Sets the constraint for the conversation optional args.

```
public void SetConstraint(ConstraintType constraintType, string constraintString)
```

Parameters

constraintType [ConstraintType](#)

The type of constraint.

constraintString string

The constraint pattern/schema/grammar string.

SetMaxOutputTokens(int)

Sets the maximum number of output tokens for the conversation optional arguments.

```
public void SetMaxOutputTokens(int maxOutputTokens)
```

Parameters

maxOutputTokens int

The maximum number of output tokens.

SetNoRepeatNgramConfig(NoRepeatNgramConfig?)

Sets the no repeat ngram configuration for the per-turn conversation optional arguments.

```
public void SetNoRepeatNgramConfig(NoRepeatNgramConfig? noRepeatNgramConfig)
```

Parameters

noRepeatNgramConfig [NoRepeatNgramConfig](#)

The no repeat ngram configuration. The contents are deep-copied when set. If [null](#), clears any previously set no repeat ngram config so no repeat ngram banning applies.

Remarks

The configured parameters ([no_repeat_ngram_size](#), [window_size](#)) apply exclusively to the output sequence generated during the current [SendMessage\(string, string?, ConversationOptionalArgs?\)](#) or [SendMessageStream\(StreamCallback, string, string?, ConversationOptionalArgs?\)](#) call.

SetRepetitionPenaltyConfig(RepetitionPenaltyConfig?)

Sets the repetition penalty configuration for the per-turn conversation optional arguments.

```
public void SetRepetitionPenaltyConfig(RepetitionPenaltyConfig?
    repetitionPenaltyConfig)
```

Parameters

[repetitionPenaltyConfig](#) [RepetitionPenaltyConfig](#)

The repetition penalty configuration. The contents are deep-copied when set. If [null](#), clears any previously set repetition penalty config so no penalties apply.

Remarks

The configured penalties ([repetition_penalty](#), [presence_penalty](#), [frequency_penalty](#), [window_size](#)) apply exclusively to the output sequence generated during the current [SendMessage\(string, string?, ConversationOptionalArgs?\)](#) or [SendMessageStream\(StreamCallback, string, string?, ConversationOptionalArgs?\)](#) call.

SetSuppressTokensConfig(SuppressTokensConfig?)

Sets the suppress tokens configuration for the per-turn conversation optional arguments.

```
public void SetSuppressTokensConfig(SuppressTokensConfig? suppressTokensConfig)
```

Parameters

[suppressTokensConfig](#) [SuppressTokensConfig](#)

The suppress tokens configuration. The contents are deep-copied when set. If [null](#) or if the inner token set is disabled/empty, clears any previously set suppress tokens config so no token suppression applies.

Remarks

The configured list of suppressed tokens applies exclusively to the output sequence generated during the current [SendMessage\(string, string?, ConversationOptionalArgs?\)](#) or [SendMessageStream\(StreamCallback, string, string?, ConversationOptionalArgs?\)](#) call.

SetThinkingConfig(ThinkingConfig?)

Sets the thinking config for the conversation optional args.

```
public void SetThinkingConfig(ThinkingConfig? thinkingConfig)
```

Parameters

thinkingConfig [ThinkingConfig](#)

The thinking config to set. If [null](#), clears any previously set thinking config.

SetVisualTokenBudget(int)

Sets the visual token budget for the conversation optional arguments.

```
public void SetVisualTokenBudget(int visualTokenBudget)
```

Parameters

visualTokenBudget int

The visual token budget.

Class DetokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class DetokenizeResult : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← DetokenizeResult

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Methods

GetString()

Returns the string.

```
public string? GetString()
```

Returns

string

The detokenized string.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class Engine

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Engine : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← Engine

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

Engine(EngineSettings)

Creates a managed wrapper around a LiteRT LM engine from the given settings. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public Engine(EngineSettings settings)
```

Parameters

settings [EngineSettings](#)

The engine settings.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

CreateSession(SessionConfig?)

Creates a managed wrapper around a LiteRT LM session. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public Session? CreateSession(SessionConfig? config = null)
```

Parameters

config [SessionConfig](#)

The session configuration to use. If [null](#), the default session configuration is used.

Returns

[Session](#)

The created session wrapper, or [null](#) on failure.

Detokenize(int[])

Detokenizes token ids using the engine's tokenizer.

```
public DetokenizeResult? Detokenize(int[] tokens)
```

Parameters

tokens int[]

An array of token ids to detokenize.

Returns

[DetokenizeResult](#)

The detokenize result wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

GetStartToken()

Returns the configured start token (BOS), if any.

```
public TokenUnion? GetStartToken()
```

Returns

[TokenUnion](#)

The start token wrapper, or [null](#) if none is configured. The caller is responsible for disposing the returned wrapper.

GetStopTokens()

Returns the configured stop tokens (EOS).

```
public TokenUnions? GetStopTokens()
```

Returns

[TokenUnions](#)

The stop tokens wrapper, or [null](#) if none are configured. The caller is responsible for disposing the returned wrapper.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Tokenize(string)

Tokenizes text using the engine's tokenizer.

```
public TokenizeResult? Tokenize(string text)
```

Parameters

text string

The UTF-8 string to tokenize.

Returns

[TokenizeResult](#)

The tokenize result wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

Class EngineSettings

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class EngineSettings : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← EngineSettings

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

EngineSettings(int, string, string?, string?)

Creates managed engine settings from a raw file descriptor. The engine takes ownership of the file descriptor and closes it when done. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public EngineSettings(int fd, string backend, string? visionBackend = null, string? audioBackend = null)
```

Parameters

fd int

The file descriptor of the model.

backend string

The backend to use (for example, "cpu" or "gpu").

visionBackend string

The vision backend to use, or [null](#) if not set.

audioBackend string

The audio backend to use, or [null](#) if not set.

Remarks

See [BackendNames](#) for valid backend strings.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

EngineSettings(string, string, string?, string?)

Creates managed engine settings from a model path. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public EngineSettings(string modelPath, string backend, string? visionBackend = null, string? audioBackend = null)
```

Parameters

modelPath string

The path to the model file.

backend string

The backend to use (for example, "cpu" or "gpu").

visionBackend string

The vision backend to use, or [null](#) if not set.

audioBackend string

The audio backend to use, or [null](#) if not set.

Remarks

See [BackendNames](#) for valid backend strings.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

EnableBenchmark()

Enables benchmarking for the engine.

```
public void EnableBenchmark()
```

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetActivationDataType(ActivationDataType)

Sets the activation data type.

```
public void SetActivationDataType(ActivationDataType activationDataType)
```

Parameters

activationDataType [ActivationDataType](#)

The activation data type.

SetAudioLoraRank(int)

Sets the Audio LoRA rank for the engine.

```
public void SetAudioLoraRank(int loraRank)
```

Parameters

loraRank int

The Audio LoRA rank.

SetAudioNumThreads(int)

Sets the number of threads for the audio CPU backend.

```
public void SetAudioNumThreads(int numThreads)
```

Parameters

numThreads int

The number of threads.

SetCacheDir(string)

Sets the cache directory for the engine.

```
public void SetCacheDir(string cacheDir)
```

Parameters

cacheDir string

The cache directory.

SetEnableSpeculativeDecoding(bool)

Sets whether to enable speculative decoding.

```
public void SetEnableSpeculativeDecoding(bool enableSpeculativeDecoding)
```

Parameters

enableSpeculativeDecoding bool

Whether to enable speculative decoding.

SetEnableYNNPACK(bool)

Sets whether YNNPACK should delegate supported operations before XNNPACK.

```
public void SetEnableYNNPACK(bool enableYNNPACK)
```

Parameters

enableYNNPACK bool

Whether to enable YNNPACK.

SetGPUDecodeStepsPerSync(int)

Sets the number of decode steps per sync for the GPU backend.

```
public void SetGPUDecodeStepsPerSync(int numDecodeStepsPerSync)
```

Parameters

numDecodeStepsPerSync int

The number of decode steps per sync.

Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

SetGPUWaitForWeightUploads(bool)

Sets whether to wait for weight uploads for the GPU backend.

```
public void SetGPUWaitForWeightUploads(bool waitForWeightUploads)
```

Parameters

`waitForWeightUploads` bool

Whether to wait for weight uploads.

Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

SetLitertDispatchLibDir(string)

Sets the LiteRT dispatch library directory for the NPU backend.

```
public void SetLitertDispatchLibDir(string libDir)
```

Parameters

`libDir` string

The dispatch library directory.

SetLoraRank(int)

Sets the LoRA rank for the engine.

```
public void SetLoraRank(int loraRank)
```

Parameters

`loraRank` int

The LoRA rank.

SetMaxNumImages(int)

Sets the maximum number of images for the engine. This is only used for the legacy implementation of the engine.

```
public void SetMaxNumImages(int maxNumImages)
```

Parameters

maxNumImages int

The maximum number of images.

SetMaxNumTokens(int)

Sets the maximum number of tokens for the engine.

```
public void SetMaxNumTokens(int maxNumTokens)
```

Parameters

maxNumTokens int

The maximum number of tokens.

SetNumDecodeTokens(int)

Sets the number of decode tokens for benchmarking.

```
public void SetNumDecodeTokens(int numDecodeTokens)
```

Parameters

numDecodeTokens int

The number of decode tokens.

SetNumPrefillTokens(int)

Sets the number of prefill tokens for benchmarking.

```
public void SetNumPrefillTokens(int numPrefillTokens)
```

Parameters

numPrefillTokens int

The number of prefill tokens.

SetNumThreads(int)

Sets the number of threads for the CPU backend.

```
public void SetNumThreads(int numThreads)
```

Parameters

numThreads int

The number of threads.

SetParallelFileSectionLoading(bool)

Sets whether the engine should load different sections of the litertlm file in parallel. Defaults to [true](#).

```
public void SetParallelFileSectionLoading(bool parallelFileSectionLoading)
```

Parameters

parallelFileSectionLoading bool

Whether to load in parallel.

SetPrefillChunkSize(int)

Sets the prefill chunk size for the engine. Only applicable for the CPU backend with dynamic models.

```
public void SetPrefillChunkSize(int prefillChunkSize)
```

Parameters

prefillChunkSize int

The prefill chunk size.

SetSupportedAudioLoraRanks(int[])

Sets the supported Audio LoRA ranks for the engine.

```
public int SetSupportedAudioLoraRanks(int[] loraRanks)
```

Parameters

loraRanks int[]

An array of supported Audio LoRA ranks.

Returns

int

0 on success, non-zero on failure.

SetSupportedLoraRanks(int[])

Sets the supported LoRA ranks for the engine.

```
public int SetSupportedLoraRanks(int[] loraRanks)
```

Parameters

loraRanks int[]

An array of supported LoRA ranks.

Returns

int

0 on success, non-zero on failure.

SetUseRingbuffersLocalAttention(bool)

Sets whether to use ringbuffers for local attention KV cache.

```
public void SetUseRingbuffersLocalAttention(bool useRingbuffersLocalAttention)
```

Parameters

`useRingbuffersLocalAttention` bool

Whether to use ringbuffers for local attention.

Remarks

When enabled for supported models, a ringbuffer stores only necessary KV cache memory for local attention layers, minimizing memory usage. When disabled, memory is allocated for the full context length, enabling instant rewinding at the cost of higher memory usage.

Note: This feature is backend-agnostic in interface design, but currently only supported by the GPU Artisan backend. Enabling it on unsupported models or backends will be ignored with a warning.

Class InputData

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class InputData : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← InputData

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

InputData(string)

Creates a managed wrapper around text input data. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public InputData(string data)
```

Parameters

data string

The text data.

Exceptions

ArgumentException

Thrown if **data** is null or empty.

InvalidOperationException

Thrown if the native object could not be created.

InputData(InputDataType, nint, nuint)

Creates a managed wrapper around LiteRT LM input data. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public InputData(InputDataType dataType, nint nativeData, nuint size)
```

Parameters

dataType [InputDataType](#)

The type of the input data.

nativeData nint

The data buffer. The data is copied internally.

size nuint

The size of the data in bytes.

Exceptions

ArgumentException

Thrown if **nativeData** or **size** is invalid.

InvalidOperationException

Thrown if the native object could not be created.

InputData(InputDataType, ReadOnlySpan<byte>)

Creates a managed wrapper around binary input data. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public InputData(InputDataType dataType, ReadOnlySpan<byte> data)
```

Parameters

dataType [InputDataType](#)

The type of the input data.

`data` ReadOnlySpan<byte>

The data buffer.

Exceptions

ArgumentException

Thrown if `data` is empty.

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Enum InputDataType

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the type of input data.

```
public enum InputDataType
```

Fields

Audio = 3

AudioEnd = 4

Image = 1

ImageEnd = 2

Text = 0

A UTF-8 string.

Class JsonResponse

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class JsonResponse : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← JsonResponse

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Methods

GetString()

Returns the JSON response string.

```
public string? GetString()
```

Returns

string

The response JSON string.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class LiteRTLMNativeHandle

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public abstract class LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle

Derived

[BenchmarkInfo](#), [Conversation](#), [ConversationConfig](#), [ConversationOptionalArgs](#), [DetokenizeResult](#), [Engine](#), [EngineSettings](#), [InputData](#), [JsonResponse](#), [NoRepeatNgramConfig](#), [RepetitionPenaltyConfig](#), [Responses](#), [SamplerParams](#), [Session](#), [SessionConfig](#), [SuppressTokensConfig](#), [ThinkingConfig](#), [TokenUnion](#), [TokenUnions](#), [TokenizeResult](#)

Properties

Native

```
protected nint Native { get; init; }
```

Property Value

nint

Methods

Dispose()

Destroys the native LiteRT LM object.

```
public void Dispose()
```

~LiteRTLMLNativeHandle()

```
protected ~LiteRTLMLNativeHandle()
```

ReleaseUnmanagedResources()

```
protected abstract void ReleaseUnmanagedResources()
```

ThrowIfDisposed()

```
protected void ThrowIfDisposed()
```

Operators

implicit operator nint(LiteRTLMLNativeHandle?)

```
public static implicit operator nint(LiteRTLMLNativeHandle? handle)
```

Parameters

handle [LiteRTLMLNativeHandle](#)

Returns

nint

Class LiteRTLMNativeLogging

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public static class LiteRTLMNativeLogging
```

Inheritance

object ← LiteRTLMNativeLogging

Methods

SetMinLogLevel(LogSeverity)

Sets the minimum log level for the LiteRT LM library.

```
public static void SetMinLogLevel(LogSeverity level)
```

Parameters

level [LogSeverity](#)

Enum LogSeverity

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the log severity / level.

```
public enum LogSeverity
```

Fields

Debug = 1

Error = 4

Fatal = 5

Info = 2

Silent = 1000

Verbose = 0

Warning = 3

Class NoRepeatNgramConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class NoRepeatNgramConfig : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← NoRepeatNgramConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

NoRepeatNgramConfig()

Creates a LiteRT LM No Repeat Ngram Config with default values ([no_repeat_ngram_size](#) = 0, [window_size](#) = 0, which means no repeat ngram banning is disabled).

```
public NoRepeatNgramConfig()
```

Remarks

When [no_repeat_ngram_size](#) is set greater than 0, any sequence of tokens (an ngram of that exact length) generated during decoding or present inside the window history can only occur at most once. If generating a candidate token would complete a repeating ngram, that candidate token's logit is set to -inf.

The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetNoRepeatNgramSize(int)

Sets the no repeat ngram size for the no repeat ngram config.

```
public void SetNoRepeatNgramSize(int noRepeatNgramSize)
```

Parameters

noRepeatNgramSize int

The size of ngrams (consecutive token sequences) that are banned from repeating within the generation history window. If set > 0 , when generating the next token would complete an already observed **no_repeat_ngram_size** sequence, the logit of the candidate token is set to $-\infty$. If set ≤ 0 , no repeat ngram banning is disabled. Negative values are automatically clamped to 0 during execution.

SetWindowSize(int)

Sets the window size for the no repeat ngram config.

```
public void SetWindowSize(int windowSize)
```

Parameters

windowSize int

The maximum number of recent tokens in generation history to consider when checking for repeating ngrams. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be ≥ 0 ; negative values are clamped to 0. If **window_size** is greater than 0 but less than **no_repeat_ngram_size**, it is automatically clamped to **no_repeat_ngram_size** so that the ngrams can fit and be tracked.

Class RepetitionPenaltyConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class RepetitionPenaltyConfig : LiteRTLMNativeHandle
```

Inheritance

object $\leftarrow$ [LiteRTLMNativeHandle](#) $\leftarrow$ RepetitionPenaltyConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

RepetitionPenaltyConfig()

Creates a LiteRT LM Repetition Penalty Config with default values ([repetition_penalty](#) = 1.0f, [presence_penalty](#) = 0.0f, [frequency_penalty](#) = 0.0f, [window_size](#) = 0, which means all history with no penalties active).

```
public RepetitionPenaltyConfig()
```

Remarks

When multiple penalties are configured and active, the order of application to output logits during decoding is:

1. Multiplicative penalty ([repetition_penalty](#))
2. Subtractive penalties ([presence_penalty](#) and [frequency_penalty](#)).

The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetFrequencyPenalty(float)

Sets the subtractive frequency penalty for the repetition penalty config.

```
public void SetFrequencyPenalty(float frequencyPenalty)
```

Parameters

frequencyPenalty float

A scalar subtracted from a token's logit, scaled linearly by the number of times that token has previously appeared inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

SetPresencePenalty(float)

Sets the subtractive presence penalty for the repetition penalty config.

```
public void SetPresencePenalty(float presencePenalty)
```

Parameters

presencePenalty float

A scalar subtracted from a token's logit if that token has appeared at least once inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

SetRepetitionPenalty(float)

Sets the multiplicative repetition penalty for the repetition penalty config.

```
public void SetRepetitionPenalty(float repetitionPenalty)
```

Parameters

repetitionPenalty float

A multiplicative penalty applied to a token's logit if that token has appeared at least once inside the generated window history (e.g., 1.0 = no penalty, 1.2 = moderate penalty). Positive logits are divided by this parameter, and negative logits are multiplied (HuggingFace style). The parameter must be $\geq 1.0f$; values less than 1.0f are automatically clamped to 1.0f during execution.

SetWindowSize(int)

Sets the window size for the repetition penalty config.

```
public void SetWindowSize(int windowSize)
```

Parameters

windowSize int

The maximum number of recent tokens in generation history to consider when computing penalization. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be ≥ 0 ; negative values are clamped to 0 during execution.

Class Responses

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Responses : LiteRTLMNativeHandle
```

Inheritance

object $\leftarrow$ [LiteRTLMNativeHandle](#) $\leftarrow$ Responses

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Properties

Count

Gets the number of elements in the collection.

```
public int Count { get; }
```

Property Value

int

The number of elements in the collection.

this[int]

Gets the element at the specified index in the read-only list.

```
public Responses.Candidate this[int index] { get; }
```

Parameters

index int

The zero-based index of the element to get.

Property Value

[Responses.Candidate](#)

The element at the specified index in the read-only list.

Methods

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Responses.Candidate> GetEnumerator()
```

Returns

`IEnumerator<Responses.Candidate>`

An enumerator that can be used to iterate through the collection.

Remarks

The returned enumerator is only valid for the lifetime of this [Responses](#) object.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class Responses.Candidate

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public record Responses.Candidate
```

Inheritance

object ← Responses.Candidate

Constructors

Candidate(string?, bool, float, bool, int, bool, float[]?)

```
public Candidate(string? Text, bool HasScore, float Score, bool HasTokenLength, int  
TokenLength, bool HasTokenScores, float[]? TokenScores)
```

Parameters

Text string

The response text.

HasScore bool

Does the response have a valid [Score](#)?

Score float

The response score.

HasTokenLength bool

Does the response have a valid [TokenLength](#)?

TokenLength int

The token length of this response.

HasTokenScores bool

Does the response have a valid [TokenScores](#)?

TokenScores float[]

The token scores for this response.

Properties

HasScore

Does the response have a valid [Score](#)?

```
public bool HasScore { get; init; }
```

Property Value

bool

HasTokenLength

Does the response have a valid [TokenLength](#)?

```
public bool HasTokenLength { get; init; }
```

Property Value

bool

HasTokenScores

Does the response have a valid [TokenScores](#)?

```
public bool HasTokenScores { get; init; }
```

Property Value

bool

Score

The response score.

```
public float Score { get; init; }
```

Property Value

float

Text

The response text.

```
public string? Text { get; init; }
```

Property Value

string

TokenLength

The token length of this response.

```
public int TokenLength { get; init; }
```

Property Value

int

TokenScores

The token scores for this response.

```
public float[]? TokenScores { get; init; }
```

Property Value

float[]

Class SamplerParams

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class SamplerParams : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← SamplerParams

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

SamplerParams(SamplerType)

Creates a managed wrapper around LiteRT LM sampler parameters for a specific sampler type. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public SamplerParams(SamplerType samplerType)
```

Parameters

samplerType [SamplerType](#)

The sampler type to use.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetSeed(int)

Sets the seed.

```
public void SetSeed(int seed)
```

Parameters

seed int

SetTemperature(float)

Sets the temperature.

```
public void SetTemperature(float temperature)
```

Parameters

temperature float

SetTopK(int)

Sets the top-k value.

```
public void SetTopK(int topK)
```

Parameters

topK int

SetTopP(float)

Sets the top-p value.

```
public void SetTopP(float topP)
```

Parameters

topP float

Enum SamplerType

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the type of sampler.

```
public enum SamplerType
```

Fields

Greedy = 3

Pick the token with maximum logit (i.e., argmax).

TopK = 1

Probabilistically pick among the top k tokens.

TopP = 2

Probabilistically pick among the tokens such that the sum is greater than or equal to p tokens after first performing top-k sampling.

Class Session

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Session : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← Session

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Methods

CancelProcess()

Cancels the current processing in the session.

```
public void CancelProcess()
```

GenerateContent(ReadOnlySpan<InputData>)

Generates content from the input prompt.

```
public Responses? GenerateContent(ReadOnlySpan<InputData> inputs)
```

Parameters

inputs ReadOnlySpan<[InputData](#)>

An array of input wrappers representing multimodal input.

Returns

[Responses](#)

The responses wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

GenerateContentStream(ReadOnlySpan<InputData>, StreamCallback)

Generates content from the input prompt and streams the response through a callback. This is a non-blocking call and invokes the callback from a background thread for each chunk.

```
public int GenerateContentStream(ReadOnlySpan<InputData> inputs,
    StreamCallback callback)
```

Parameters

inputs [ReadOnlySpan<InputData>](#)

An array of input wrappers representing multimodal input.

callback [StreamCallback](#)

The callback function ([StreamCallback](#)) that receives response chunks.

Returns

int

0 on success, non-zero on failure to start the stream.

GetBenchmarkInfo()

Retrieves benchmark information for the session. The caller is responsible for disposing the returned wrapper.

```
public BenchmarkInfo? GetBenchmarkInfo()
```

Returns

[BenchmarkInfo](#)

The benchmark information wrapper, or [null](#) on failure.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

RewindToCheckpoint(string)

Rewinds the session to the given checkpoint label.

```
public int RewindToCheckpoint(string label)
```

Parameters

label string

Label of the checkpoint to rewind to.

Returns

int

0 on success, non-zero on failure.

RewindToStep(int)

Rewinds the session to a specific step number.

```
public int RewindToStep(int step)
```

Parameters

step int

The step number to rewind to.

Returns

int

0 on success, non-zero on failure.

RunDecode()

Starts decoding for the model to predict a response based on the input prompt or query added after calling [RunPrefill\(ReadOnlySpan<InputData>\)](#). This is a blocking call and returns when decoding is complete.

```
public Responses? RunDecode()
```

Returns

[Responses](#)

The responses wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

RunDecodeAsync(StreamCallback)

Starts decoding for the model to predict a response based on the input prompt or query added after calling [RunPrefill\(ReadOnlySpan<InputData>\)](#). This is a non-blocking call that streams response chunks through a callback.

```
public int RunDecodeAsync(StreamCallback callback)
```

Parameters

callback [StreamCallback](#)

The callback function ([StreamCallback](#)) that receives response chunks.

Returns

int

0 on success, non-zero on failure.

RunPrefill(ReadOnlySpan<InputData>)

Adds the input prompt or query to the model and starts the prefill process. This is a blocking call and returns when prefill is complete.

```
public int RunPrefill(ReadOnlySpan<InputData> inputs)
```

Parameters

inputs ReadOnlySpan<[InputData](#)>

An array of input wrappers representing multimodal input.

Returns

int

0 on success, non-zero on failure.

RunTextScoring(string[], bool)

Scores the target text after the prefill process is complete.

```
public Responses? RunTextScoring(string[] targetText, bool storeTokenLengths)
```

Parameters

targetText string[]

An array of target text strings to score.

storeTokenLengths bool

Whether to store the token lengths of the target texts in the responses.

Returns

[Responses](#)

The responses wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

SaveCheckpoint(string)

Saves the current state of the session to a checkpoint with the given label.

```
public int SaveCheckpoint(string label)
```

Parameters

label string

Label for the checkpoint.

Returns

int

0 on success, non-zero on failure.

Class SessionConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class SessionConfig : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← SessionConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

SessionConfig()

Creates a managed wrapper around a LiteRT LM session configuration. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public SessionConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetApplyPromptTemplate(bool)

Sets whether to apply the prompt template for this session.

```
public void SetApplyPromptTemplate(bool applyPromptTemplate)
```

Parameters

applyPromptTemplate bool

Whether to apply the prompt template.

SetAudioLoraPath(string)

Sets the path to the audio LoRA weights file.

```
public int SetAudioLoraPath(string audioLoraPath)
```

Parameters

audioLoraPath string

The path to the audio LoRA weights file.

Returns

int

0 on success, non-zero on failure.

SetLoraPath(string)

Sets the path to the LoRA weights file.

```
public int SetLoraPath(string loraPath)
```

Parameters

loraPath string

The path to the text LoRA weights file.

Returns

int

0 on success, non-zero on failure.

SetMaxOutputTokens(int)

Sets the maximum number of output tokens per decode step for this session.

```
public void SetMaxOutputTokens(int maxOutputTokens)
```

Parameters

maxOutputTokens int

The maximum number of output tokens.

SetSamplerParams(SamplerParams)

Sets the sampler parameters for this session configuration.

```
public void SetSamplerParams(SamplerParams samplerParams)
```

Parameters

samplerParams [SamplerParams](#)

The sampler parameters to use.

Delegate StreamCallback

Namespace: [Uralstech.UAI.LiteRTLM](#)

Callback for streaming responses.

```
public delegate void StreamCallback(StreamChunk chunk)
```

Parameters

chunk [StreamChunk](#)

The stream chunk object. It's only valid for the duration of the call.

Class StreamChunk

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class StreamChunk
```

Inheritance

object ← StreamChunk

Constructors

StreamChunk(nint)

```
public StreamChunk(nint native)
```

Parameters

native nint

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

GetError()

Gets the error message associated with this chunk, if any.

```
public string? GetError()
```

Returns

string

Remarks

Returns [null](#) if there is no error.

GetText()

Gets the text content of the chunk.

```
public string? GetText()
```

Returns

string

Returns [null](#) if there is no text content in this chunk (e.g. if it is an error or metadata-only chunk).

IsFinal()

Returns [true](#) if this is the final chunk of the stream.

```
public bool IsFinal()
```

Returns

bool

Operators

implicit operator nint(StreamChunk?)

```
public static implicit operator nint(StreamChunk? handle)
```

Parameters

`handle` [StreamChunk](#)

Returns

`nint`

Class SuppressTokensConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class SuppressTokensConfig : LiteRTLMNativeHandle
```

Inheritance

object $\leftarrow$ [LiteRTLMNativeHandle](#) $\leftarrow$ SuppressTokensConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

SuppressTokensConfig()

Creates a LiteRT LM Suppress Tokens Config with default values (an empty set of suppressed tokens, which means token suppression is disabled).

```
public SuppressTokensConfig()
```

Remarks

When `suppress_tokens` is configured with one or more token IDs, the logits corresponding to those exact token IDs will be set directly to -inf during generation. This guarantees that those tokens can never be sampled by the model.

The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetSuppressTokens(int[]?)

Sets the list of token IDs to suppress for the suppress tokens config.

```
public void SetSuppressTokens(int[]? suppressTokens)
```

Parameters

suppressTokens int[]

An array of integer token IDs that should be banned from generation. During every decode step, each listed token ID's candidate logit will be forced to -inf. If **suppress_tokens** is [null](#) or empty, any previously set suppressed tokens are cleared and token suppression is disabled.

Class ThinkingConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class ThinkingConfig : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← ThinkingConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Constructors

ThinkingConfig()

Creates a managed wrapper around a LiteRT LM thinking configuration. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public ThinkingConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetEnableThinking(bool)

Sets whether thinking/reasoning generation is enabled.

```
public void SetEnableThinking(bool enableThinking)
```

Parameters

enableThinking bool

Whether thinking is enabled.

SetThinkingTokenBudget(int)

Sets the thinking token budget.

```
public void SetThinkingTokenBudget(int thinkingTokenBudget)
```

Parameters

thinkingTokenBudget int

Budget for token-by-token reasoning generation (-1 for infinite).

Class TokenUnion

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class TokenUnion : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← TokenUnion

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Methods

GetIds(out int[]?)

Returns the token ids from a token union.

```
public int GetIds(out int[]? tokenIds)
```

Parameters

tokenIds int[]

The token IDs.

Returns

int

0 on success, non-zero if the token union does not contain token ids.

GetString()

Returns the string value from a token union.

```
public string? GetString()
```

Returns

string

The string value, or [null](#) if the token union does not contain a string value.

GetUnionType()

Returns the type of the token union.

```
public TokenUnionType GetUnionType()
```

Returns

[TokenUnionType](#)

The type of the token union.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Enum TokenUnionType

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the type of a TokenUnion.

```
public enum TokenUnionType
```

Fields

`Ids = 1`

`String = 0`

Class TokenUnions

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class TokenUnions : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← TokenUnions

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Methods

GetTokenUnions()

Gets the token unions in this collection.

```
public TokenUnion[] GetTokenUnions()
```

Returns

[TokenUnion](#)[]

The token union wrappers in the collection. The caller is responsible for disposing each returned wrapper.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class TokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class TokenizeResult : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← TokenizeResult

Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

Properties

Length

The number of token ids from the tokenize result.

```
public int Length { get; }
```

Property Value

int

Methods

AsReadOnlySpan()

Returns a span of the token ids.

```
public ReadOnlySpan<int> AsReadOnlySpan()
```

Returns

ReadOnlySpan<int>

Remarks

The returned span is only valid for the lifetime of this [TokenizeResult](#) object.

CopyTo(Span<int>)

Copies the token ids into `span`.

```
public long CopyTo(Span<int> span)
```

Parameters

`span` Span<int>

Returns

long

The number of copied elements.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Namespace Uralstech.UAI.LiteRTLM.Native

Classes

[NativeAPI](#)

[NativeAPI.BenchmarkInfo](#)

[NativeAPI.Conversation](#)

[NativeAPI.ConversationConfig](#)

[NativeAPI.ConversationOptionalArgs](#)

[NativeAPI.DetokenizeResult](#)

[NativeAPI.Engine](#)

[NativeAPI.EngineSettings](#)

[NativeAPI.InputData](#)

[NativeAPI.JsonResponse](#)

[NativeAPI.NoRepeatNgramConfig](#)

[NativeAPI.RepetitionPenaltyConfig](#)

[NativeAPI.Responses](#)

[NativeAPI.SamplerParams](#)

[NativeAPI.Session](#)

[NativeAPI.SessionConfig](#)

[NativeAPI.StreamChunk](#)

[NativeAPI.SuppressTokensConfig](#)

[NativeAPI.ThinkingConfig](#)

[NativeAPI.TokenUnion](#)

[NativeAPI.TokenUnions](#)

[NativeAPI.TokenizeResult](#)

Delegates

[NativeAPI.StreamCallback](#)

Callback for streaming responses.

Class NativeAPI

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI
```

Inheritance

object ← NativeAPI

Fields

LibLiteRTLM

```
public const string LibLiteRTLM = "litert-lm"
```

Field Value

string

Methods

litert_lm_set_min_log_level(LogSeverity)

Sets the minimum log level for the LiteRT LM library.

```
public static extern void litert_lm_set_min_log_level(LogSeverity level)
```

Parameters

level [LogSeverity](#)

Class NativeAPI.BenchmarkInfo

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.BenchmarkInfo
```

Inheritance

object ← NativeAPI.BenchmarkInfo

Methods

litert_lm_benchmark_info_delete(nint)

Destroys a LiteRT LM Benchmark Info object.

```
public static extern void litert_lm_benchmark_info_delete(nint benchmarkInfo)
```

Parameters

benchmarkInfo nint

The benchmark info to destroy.

litert_lm_benchmark_info_get_decode_token_count_at(nint, int)

Returns the decode token count at a given turn index.

```
public static extern int litert_lm_benchmark_info_get_decode_token_count_at(nint benchmarkInfo, int index)
```

Parameters

benchmarkInfo nint

The benchmark info object.

index int

The index of the decode turn.

Returns

int

The decode token count.

litert_lm_benchmark_info_get_decode_tokens_per_sec_at(nint, int)

Returns the decode tokens per second at a given turn index.

```
public static extern double  
litert_lm_benchmark_info_get_decode_tokens_per_sec_at(nint benchmarkInfo, int index)
```

Parameters

benchmarkInfo nint

The benchmark info object.

index int

The index of the decode turn.

Returns

double

The decode tokens per second.

litert_lm_benchmark_info_get_num_decode_turns(nint)

Returns the number of decode turns.

```
public static extern int litert_lm_benchmark_info_get_num_decode_turns(nint  
benchmarkInfo)
```

Parameters

benchmarkInfo nint

The benchmark info object.

Returns

int

The number of decode turns.

litert_lm_benchmark_info_get_num_prefill_turns(nint)

Returns the number of prefill turns.

```
public static extern int litert_lm_benchmark_info_get_num_prefill_turns(nint benchmarkInfo)
```

Parameters

benchmarkInfo nint

The benchmark info object.

Returns

int

The number of prefill turns.

litert_lm_benchmark_info_get_prefill_token_count_at(nint, int)

Returns the prefill token count at a given turn index.

```
public static extern int litert_lm_benchmark_info_get_prefill_token_count_at(nint benchmarkInfo, int index)
```

Parameters

benchmarkInfo nint

The benchmark info object.

index int

The index of the prefill turn.

Returns

int

The prefill token count.

litert_lm_benchmark_info_get_prefill_tokens_per_sec_at(nint, int)

Returns the prefill tokens per second at a given turn index.

```
public static extern double  
litert_lm_benchmark_info_get_prefill_tokens_per_sec_at(nint benchmarkInfo,  
int index)
```

Parameters

benchmarkInfo nint

The benchmark info object.

index int

The index of the prefill turn.

Returns

double

The prefill tokens per second.

litert_lm_benchmark_info_get_time_to_first_token(nint)

Returns the time to the first token in seconds.

```
public static extern double litert_lm_benchmark_info_get_time_to_first_token(nint benchmarkInfo)
```

Parameters

benchmarkInfo nint

The benchmark info object.

Returns

double

The time to the first token in seconds.

Remarks

Note that the first time to token doesn't include the time for initialization. It is the sum of the prefill time for the first turn and the time spent for decoding the first token.

litert_lm_benchmark_info_get_total_init_time_in_second(nint)

Returns the total initialization time in seconds.

```
public static extern double  
litert_lm_benchmark_info_get_total_init_time_in_second(nint benchmarkInfo)
```

Parameters

benchmarkInfo nint

The benchmark info object.

Returns

double

The total initialization time in seconds.

Class NativeAPI.Conversation

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Conversation
```

Inheritance

object ← NativeAPI.Conversation

Methods

litert_lm_conversation_cancel_process(nint)

Cancels the ongoing inference process, for asynchronous inference.

```
public static extern void litert_lm_conversation_cancel_process(nint conversation)
```

Parameters

conversation nint

The conversation to cancel the inference for.

litert_lm_conversation_clone(nint)

Clones a LiteRT LM Conversation, duplicating its prefilled state. The caller is responsible for destroying the cloned conversation using [litert_lm_conversation_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_clone(nint conversation)
```

Parameters

conversation nint

The conversation to clone.

Returns

nint

A pointer to the cloned conversation, or nint.Zero on failure.

litert_lm_conversation_create(nint, nint)

Creates a LiteRT LM Conversation. The caller is responsible for destroying the conversation using [litert_lm_conversation_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_create(nint engine, nint config)
```

Parameters

engine nint

The engine to create the conversation from.

config nint

The conversation config to use. If nint.Zero, the default config will be used.

Returns

nint

A pointer to the created conversation, or nint.Zero on failure.

litert_lm_conversation_delete(nint)

Destroys a LiteRT LM Conversation.

```
public static extern void litert_lm_conversation_delete(nint conversation)
```

Parameters

conversation nint

The conversation to destroy.

litert_lm_conversation_get_benchmark_info(nint)

Retrieves the benchmark information from the conversation. The caller is responsible for destroying the benchmark info using [litert_lm_benchmark_info_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_get_benchmark_info(nint
conversation)
```

Parameters

conversation nint

The conversation to get the benchmark info from.

Returns

nint

A pointer to the benchmark info, or nint.Zero on failure.

litert_lm_conversation_get_token_count(nint)

Gets the number of tokens in the conversation KV Cache (prefill + decode).

```
public static extern int litert_lm_conversation_get_token_count(nint conversation)
```

Parameters

conversation nint

Returns

int

The number of tokens, or a negative value on failure.

litert_lm_conversation_render_message_to_string(nint, string)

Renders the message into a string according to the template.

```
public static extern nint litert_lm_conversation_render_message_to_string(nint
conversation, string messageJson)
```

Parameters

conversation nint

The conversation instance.

messageJson string

A JSON string representing the message to render.

Returns

nint

A pointer to the rendered string, or nint.Zero on failure. The returned string is owned by the Conversation object and is valid until the next call to this function or until the conversation is deleted.

Remarks

This function does not need to be called for actual message sending, as the [litert_lm_conversation_send_message\(nint, string, string?, nint\)](#) and [litert_lm_conversation_send_message_stream\(nint, string, string?, nint, nint, nint\)](#) functions will handle rendering internally.

litert_lm_conversation_render_preface_to_string(nint)

Renders the preface into a string according to the template.

```
public static extern nint litert_lm_conversation_render_preface_to_string(nint
conversation)
```

Parameters

conversation nint

The conversation instance.

Returns

nint

A pointer to the rendered string, or nint.Zero on failure. The returned string is owned by the Conversation object and is valid until the next call to this function or until the conversation is deleted.

litert_lm_conversation_send_message(nint, string, string?, nint)

Sends a message to the conversation and returns the response. This is a blocking call.

```
public static extern nint litert_lm_conversation_send_message(nint conversation,
string messageJson, string? extraContext, nint optionalArgs)
```

Parameters

conversation nint

The conversation to use.

messageJson string

A JSON string representing the message to send.

extraContext string

A JSON string representing the extra context to use.

optionalArgs nint

A pointer to the optional arguments to use.

Returns

nint

A pointer to the JSON Response, or nint.Zero on failure. The caller is responsible for deleting the response using [litert_lm_json_response_delete\(nint\)](#).

litert_lm_conversation_send_message_stream(nint, string, string?, nint, nint, nint)

Sends a message to the conversation and streams the response via a callback. This is a non-blocking call that will invoke the callback from a background thread for each chunk.

```
public static extern int litert_lm_conversation_send_message_stream(nint conversation, string messageJson, string? extraContext, nint optionalArgs, nint callback, nint callbackData)
```

Parameters

conversation nint

The conversation to use.

messageJson string

A JSON string representing the message to send.

extraContext string

A JSON string representing the extra context to use.

optionalArgs nint

A pointer to the optional arguments to use.

callback nint

The callback function ([NativeAPI.StreamCallback](#)) to receive response chunks.

callbackData nint

A pointer to user data that will be passed to the callback.

Returns

int

0 on success, non-zero on failure to start the stream.

Class NativeAPI.ConversationConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.ConversationConfig
```

Inheritance

object ← NativeAPI.ConversationConfig

Methods

litert_lm_conversation_config_create()

Creates a LiteRT LM Conversation Config. The caller is responsible for destroying the config using [litert_lm_conversation_config_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_conversation_config_delete(nint)

Destroys a LiteRT LM Conversation Config.

```
public static extern void litert_lm_conversation_config_delete(nint config)
```

Parameters

config nint

The config to destroy.

litert_lm_conversation_config_set_constraint_provider(nint, nint)

Sets the constraint provider type for this conversation config.

```
public static extern void litert_lm_conversation_config_set_constraint_provider(nint config, nint providerType)
```

Parameters

config nint

The config to modify.

providerType nint

A pointer to the constraint provider type to use ([ConstraintProviderType](#)), or nint.Zero to unset.

litert_lm_conversation_config_set_enable_constrained_decoding(nint, bool)

Sets whether to enable constrained decoding for this conversation config.

```
public static extern void  
litert_lm_conversation_config_set_enable_constrained_decoding(nint config,  
bool enableConstrainedDecoding)
```

Parameters

config nint

The config to modify.

enableConstrainedDecoding bool

Whether to enable constrained decoding.

litert_lm_conversation_config_set_extra_context(nint, string)

Sets the extra context for the conversation preface.

```
public static extern void litert_lm_conversation_config_set_extra_context(nint config, string extraContextJson)
```

Parameters

config nint

The config to modify.

extraContextJson string

A JSON string representing the extra context object.

litert_lm_conversation_config_set_filter_channel_content_from_kv_cache(nint, bool)

Sets whether to filter channel content from the KV cache.

```
public static extern void litert_lm_conversation_config_set_filter_channel_content_from_kv_cache(nint config, bool filterChannelContentFromKvCache)
```

Parameters

config nint

The config to modify.

filterChannelContentFromKvCache bool

Whether to filter channel content.

litert_lm_conversation_config_set_messages(nint, string)

Sets the initial messages for this conversation config.

```
public static extern void litert_lm_conversation_config_set_messages(nint config,  
string messagesJson)
```

Parameters

config nint

The config to modify.

messagesJson string

The initial messages in JSON array format.

litert_lm_conversation_config_set_prompt_template(nint, string)

Sets the prompt template for this conversation config.

```
public static extern void litert_lm_conversation_config_set_prompt_template(nint  
config, string promptTemplate)
```

Parameters

config nint

The config to modify.

promptTemplate string

The prompt template string (e.g. Jinja template). If not set, use the default provided by the model or the engine.

litert_lm_conversation_config_set_session_config(nint, nint)

Sets the session config for this conversation config.

```
public static extern void litert_lm_conversation_config_set_session_config(nint
config, nint sessionConfig)
```

Parameters

config nint

The config to modify.

sessionConfig nint

The session config to use.

litert_lm_conversation_config_set_stream_tool_calls(nint, bool, string)

Sets whether to stream tool call tokens.

```
public static extern void litert_lm_conversation_config_set_stream_tool_calls(nint
config, bool streamToolCalls, string channelName)
```

Parameters

config nint

The config to modify.

streamToolCalls bool

Whether to stream tool call tokens.

channelName string

The channel name to use for tool call tokens.

litert_lm_conversation_config_set_system_message(nint, string)

Sets the system message for this conversation config.

```
public static extern void litert_lm_conversation_config_set_system_message(nint
config, string systemMessageJson)
```

Parameters

config nint

The config to modify.

systemMessageJson string

The system message in JSON format.

litert_lm_conversation_config_set_thinking_config(nint, nint)

Sets the thinking config for this conversation config.

```
public static extern void litert_lm_conversation_config_set_thinking_config(nint
config, nint thinkingConfig)
```

Parameters

config nint

The config to modify.

thinkingConfig nint

The thinking config to set. If nint.Zero, clears any previously set thinking config.

litert_lm_conversation_config_set_tools(nint, string)

Sets the tools for this conversation config.

```
public static extern void litert_lm_conversation_config_set_tools(nint config,
string toolsJson)
```

Parameters

`config` nint

The config to modify.

`toolsJson` string

The tools description in JSON array format.

Class NativeAPI.ConversationOptionalArgs

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.ConversationOptionalArgs
```

Inheritance

object ← NativeAPI.ConversationOptionalArgs

Methods

litert_lm_conversation_optional_args_create()

Creates a LiteRT LM Conversation Optional Args. The caller is responsible for destroying the optional args using [litert_lm_conversation_optional_args_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_optional_args_create()
```

Returns

nint

A pointer to the created optional args, or nint.Zero on failure.

litert_lm_conversation_optional_args_delete(nint)

Destroys a LiteRT LM Conversation Optional Args.

```
public static extern void litert_lm_conversation_optional_args_delete(nint optionalArgs)
```

Parameters

optionalArgs nint

The optional args to destroy.

litert_lm_conversation_optional_args_set_constraint(nint, ConstraintType, string)

Sets the constraint for the conversation optional args.

```
public static extern void litert_lm_conversation_optional_args_set_constraint(nint optionalArgs, ConstraintType constraintType, string constraintString)
```

Parameters

optionalArgs nint

The optional args to modify.

constraintType [ConstraintType](#)

The type of constraint.

constraintString string

The constraint pattern/schema/grammar string.

litert_lm_conversation_optional_args_set_max_output_tokens(nint, int)

Sets the maximum number of output tokens for the conversation optional args.

```
public static extern void  
litert_lm_conversation_optional_args_set_max_output_tokens(nint optionalArgs,  
int maxOutputTokens)
```

Parameters

optionalArgs nint

The optional args to modify.

maxOutputTokens int

The maximum number of output tokens.

litert_lm_conversation_optional_args_set_no_repeat_ngram_config(nint, nint)

Sets the no repeat ngram configuration for the per-turn conversation optional arguments ([OptionalArgs](#)).

```
public static extern void
litert_lm_conversation_optional_args_set_no_repeat_ngram_config(nint optionalArgs,
nint noRepeatNgramConfig)
```

Parameters

[optionalArgs](#) nint

The optional arguments structure ([OptionalArgs](#)) to modify.

[noRepeatNgramConfig](#) nint

The no repeat ngram configuration struct ([LiteRtLmNoRepeatNgramConfig](#)) created via [litert_lm_no_repeat_ngram_config_create\(\)](#). The contents are deep-copied when set. If nint. Zero, clears any previously set no repeat ngram config so no repeat ngram banning applies.

Remarks

The configured parameters ([no_repeat_ngram_size](#), [window_size](#)) apply exclusively to the output sequence generated during the current [send_message](#) or [send_message_async](#) call.

litert_lm_conversation_optional_args_set_repetition_penalty_config(nint, nint)

Sets the repetition penalty configuration for the per-turn conversation optional arguments ([OptionalArgs](#)).

```
public static extern void
litert_lm_conversation_optional_args_set_repetition_penalty_config(nint
optionalArgs, nint repetitionPenaltyConfig)
```

Parameters

`optionalArgs` `nint`

The optional arguments structure (`OptionalArgs`) to modify.

`repetitionPenaltyConfig` `nint`

The repetition penalty configuration struct (`LiteRtLmRepetitionPenaltyConfig`) created via [litert_lm_repetition_penalty_config_create\(\)](#). The contents are deep-copied when set. If `nint.Zero`, clears any previously set repetition penalty config so no penalties apply.

Remarks

The configured penalties (`repetition_penalty`, `presence_penalty`, `frequency_penalty`, `window_size`) apply exclusively to the output sequence generated during the current `send_message` or `send_message_async` call.

litert_lm_conversation_optional_args_set_suppress_tokens_config(nint, nint)

Sets the suppress tokens configuration for the per-turn conversation optional arguments (`OptionalArgs`).

```
public static extern void
litert_lm_conversation_optional_args_set_suppress_tokens_config(nint optionalArgs,
nint suppressTokensConfig)
```

Parameters

`optionalArgs` `nint`

The optional arguments structure (`OptionalArgs`) to modify.

`suppressTokensConfig` `nint`

The suppress tokens configuration struct (`LiteRtLmSuppressTokensConfig`) created via [litert_lm_suppress_tokens_config_create\(\)](#). The contents are deep-copied when set. If `nint.Zero` or if the inner token set is disabled/empty, clears any previously set suppress tokens config so no token suppression applies.

Remarks

The configured list of suppressed tokens applies exclusively to the output sequence generated during the current `send_message` or `send_message_async` call.

litert_lm_conversation_optional_args_set_thinking_config(nint, nint)

Sets the thinking config for the conversation optional args.

```
public static extern void  
litert_lm_conversation_optional_args_set_thinking_config(nint optionalArgs,  
nint thinkingConfig)
```

Parameters

`optionalArgs` nint

The optional args to modify.

`thinkingConfig` nint

The thinking config to set. If `nint.Zero`, clears any previously set thinking config.

litert_lm_conversation_optional_args_set_visual_token_budget(nint, int)

Sets the visual token budget for the conversation optional args.

```
public static extern void  
litert_lm_conversation_optional_args_set_visual_token_budget(nint optionalArgs,  
int visualTokenBudget)
```

Parameters

`optionalArgs` nint

The optional args to modify.

`visualTokenBudget` int

The visual token budget.

Class NativeAPI.DetokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.DetokenizeResult
```

Inheritance

object ← NativeAPI.DetokenizeResult

Methods

litert_lm_detokenize_result_delete(nint)

Destroys a LiteRT LM Detokenize Result.

```
public static extern void litert_lm_detokenize_result_delete(nint result)
```

Parameters

result nint

The detokenize result to destroy.

litert_lm_detokenize_result_get_string(nint)

Returns the string from a detokenize result.

```
public static extern nint litert_lm_detokenize_result_get_string(nint result)
```

Parameters

result nint

The detokenize result.

Returns

nint

The detokenized string. The returned string is owned by the DetokenizeResult object and is valid only for its lifetime.

Class NativeAPI.Engine

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Engine
```

Inheritance

object ← NativeAPI.Engine

Methods

litert_lm_engine_create(nint)

Creates a LiteRT LM Engine from the given settings. The caller is responsible for destroying the engine using [litert_lm_engine_delete\(nint\)](#).

```
public static extern nint litert_lm_engine_create(nint settings)
```

Parameters

settings nint

The engine settings.

Returns

nint

A pointer to the created engine, or nint.Zero on failure.

litert_lm_engine_create_session(nint, nint)

Creates a LiteRT LM Session. The caller is responsible for destroying the session using [litert_lm_session_delete\(nint\)](#).

```
public static extern nint litert_lm_engine_create_session(nint engine, nint config)
```

Parameters

engine nint

The engine to create the session from.

config nint

The session config of the session. If nint.Zero, use the default session config.

Returns

nint

A pointer to the created session, or nint.Zero on failure.

litert_lm_engine_delete(nint)

Destroys a LiteRT LM Engine.

```
public static extern void litert_lm_engine_delete(nint engine)
```

Parameters

engine nint

The engine to destroy.

litert_lm_engine_detokenize(nint, int[], nuint)

Detokenizes token ids using the engine's tokenizer.

```
public static extern nint litert_lm_engine_detokenize(nint engine, int[] tokens, nuint numTokens)
```

Parameters

engine nint

The engine instance.

tokens int[]

An array of token ids to detokenize.

numTokens nuint

The number of token ids in the array.

Returns

nint

A pointer to the detokenize result, or nint.Zero on failure. The caller is responsible for deleting the result using [litert_lm_detokenize_result_delete\(nint\)](#).

litert_lm_engine_get_start_token(nint)

Returns the configured start token (BOS), if any.

```
public static extern nint litert_lm_engine_get_start_token(nint engine)
```

Parameters

engine nint

The engine instance.

Returns

nint

A pointer to the start token, or nint.Zero if none configured. The caller is responsible for deleting the result using [litert_lm_token_union_delete\(nint\)](#).

litert_lm_engine_get_stop_tokens(nint)

Returns the configured stop tokens (EOS).

```
public static extern nint litert_lm_engine_get_stop_tokens(nint engine)
```

Parameters

engine nint

The engine instance.

Returns

nint

A pointer to the stop tokens collection, or nint.Zero if none configured. The caller is responsible for deleting the result using [litert_lm_token_unions_delete\(nint\)](#).

litert_lm_engine_tokenize(nint, string)

Tokenizes text using the engine's tokenizer.

```
public static extern nint litert_lm_engine_tokenize(nint engine, string text)
```

Parameters

engine nint

The engine instance.

text string

The UTF-8 string to tokenize.

Returns

nint

A pointer to the tokenize result, or nint.Zero on failure. The caller is responsible for deleting the result using [litert_lm_tokenize_result_delete\(nint\)](#).

Class NativeAPI.EngineSettings

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.EngineSettings
```

Inheritance

object ← NativeAPI.EngineSettings

Methods

litert_lm_engine_settings_create(string, string, string?, string?)

Creates LiteRT LM Engine Settings. The caller is responsible for destroying the settings using [litert_lm_engine_settings_delete\(nint\)](#).

```
public static extern nint litert_lm_engine_settings_create(string modelPath, string backendStr, string? visionBackendStr, string? audioBackendStr)
```

Parameters

modelPath string

The path to the model file.

backendStr string

The backend to use (e.g., "cpu", "gpu").

visionBackendStr string

The vision backend to use, or [null](#) if not set.

audioBackendStr string

The audio backend to use, or [null](#) if not set.

Returns

nint

A pointer to the created settings, or nint.Zero on failure.

litert_lm_engine_settings_create_from_raw_file_descriptor(int, string, string?, string?)

Creates LiteRT LM Engine Settings from a raw file descriptor. The engine takes ownership of the file descriptor and will close it when done. The caller is responsible for destroying the settings using [litert_lm_engine_settings_delete\(nint\)](#).

```
public static extern nint  
litert_lm_engine_settings_create_from_raw_file_descriptor(int fd, string backendStr,  
string? visionBackendStr, string? audioBackendStr)
```

Parameters

fd int

The file descriptor of the model.

backendStr string

The backend to use (e.g., "cpu", "gpu").

visionBackendStr string

The vision backend to use, or [null](#) if not set.

audioBackendStr string

The audio backend to use, or [null](#) if not set.

Returns

nint

A pointer to the created settings, or nint.Zero on failure.

litert_lm_engine_settings_delete(nint)

Destroys LiteRT LM Engine Settings.

```
public static extern void litert_lm_engine_settings_delete(nint settings)
```

Parameters

settings nint

The settings to destroy.

litert_lm_engine_settings_enable_benchmark(nint)

Enables benchmarking for the engine.

```
public static extern void litert_lm_engine_settings_enable_benchmark(nint settings)
```

Parameters

settings nint

The engine settings.

litert_lm_engine_settings_set_activation_data_type(nint, ActivationDataType)

Sets the activation data type.

```
public static extern void litert_lm_engine_settings_set_activation_data_type(nint settings, ActivationDataType activationDataType)
```

Parameters

settings nint

The engine settings.

activationDataType [ActivationDataType](#)

The activation data type.

litert_lm_engine_settings_set_audio_lora_rank(nint, int)

Sets the Audio LoRA rank for the engine.

```
public static extern void litert_lm_engine_settings_set_audio_lora_rank(nint
settings, int loraRank)
```

Parameters

settings nint

The engine settings.

loraRank int

The Audio LoRA rank.

litert_lm_engine_settings_set_audio_num_threads(nint, int)

Sets the number of threads for the audio CPU backend.

```
public static extern void litert_lm_engine_settings_set_audio_num_threads(nint
settings, int numThreads)
```

Parameters

settings nint

The engine settings.

numThreads int

The number of threads.

litert_lm_engine_settings_set_cache_dir(nint, string)

Sets the cache directory for the engine.

```
public static extern void litert_lm_engine_settings_set_cache_dir(nint settings,  
string cacheDir)
```

Parameters

settings nint

The engine settings.

cacheDir string

The cache directory.

litert_lm_engine_settings_set_enable_speculative_decoding(nint, bool)

Sets whether to enable speculative decoding.

```
public static extern void  
litert_lm_engine_settings_set_enable_speculative_decoding(nint settings,  
bool enableSpeculativeDecoding)
```

Parameters

settings nint

The engine settings.

enableSpeculativeDecoding bool

Whether to enable speculative decoding.

litert_lm_engine_settings_set_enable_ynnpack(nint, bool)

Sets whether YNNPACK should delegate supported operations before XNNPACK.

```
public static extern void litert_lm_engine_settings_set_enable_ynnpack(nint
settings, bool enableYnnpack)
```

Parameters

settings nint

The engine settings.

enableYnnpack bool

Whether to enable YNNPACK.

litert_lm_engine_settings_set_gpu_decode_steps_per_sync(nint, int)

Sets the number of decode steps per sync for the GPU backend.

```
public static extern void
litert_lm_engine_settings_set_gpu_decode_steps_per_sync(nint settings,
int numDecodeStepsPerSync)
```

Parameters

settings nint

The engine settings.

numDecodeStepsPerSync int

The number of decode steps per sync.

Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

litert_lm_engine_settings_set_gpu_wait_for_weight_uploads(nint, bool)

Sets whether to wait for weight uploads for the GPU backend.

```
public static extern void  
litert_lm_engine_settings_set_gpu_wait_for_weight_uploads(nint settings,  
bool waitForWeightUploads)
```

Parameters

settings nint

The engine settings.

waitForWeightUploads bool

Whether to wait for weight uploads.

Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

litert_lm_engine_settings_set_litert_dispatch_lib_dir(nint, string)

Sets the LiteRT dispatch library directory for NPU backend.

```
public static extern void litert_lm_engine_settings_set_litert_dispatch_lib_dir(nint  
settings, string libDir)
```

Parameters

settings nint

The engine settings.

libDir string

The dispatch library directory.

litert_lm_engine_settings_set_lora_rank(nint, int)

Sets the LoRA rank for the engine.

```
public static extern void litert_lm_engine_settings_set_lora_rank(nint settings,  
int loraRank)
```

Parameters

settings nint

The engine settings.

loraRank int

The LoRA rank.

litert_lm_engine_settings_set_max_num_images(nint, int)

Sets the maximum number of images for the engine. This is only used for the legacy implementation of the engine.

```
public static extern void litert_lm_engine_settings_set_max_num_images(nint  
settings, int maxNumImages)
```

Parameters

settings nint

The engine settings.

maxNumImages int

The maximum number of images.

litert_lm_engine_settings_set_max_num_tokens(nint, int)

Sets the maximum number of tokens for the engine.

```
public static extern void litert_lm_engine_settings_set_max_num_tokens(nint
settings, int maxNumTokens)
```

Parameters

settings nint

The engine settings.

maxNumTokens int

The maximum number of tokens.

litert_lm_engine_settings_set_num_decode_tokens(nint, int)

Sets the number of decode tokens for benchmarking.

```
public static extern void litert_lm_engine_settings_set_num_decode_tokens(nint
settings, int numDecodeTokens)
```

Parameters

settings nint

The engine settings.

numDecodeTokens int

The number of decode tokens.

litert_lm_engine_settings_set_num_prefill_tokens(nint, int)

Sets the number of prefill tokens for benchmarking.

```
public static extern void litert_lm_engine_settings_set_num_prefill_tokens(nint
settings, int numPrefillTokens)
```

Parameters

settings nint

The engine settings.

numPrefillTokens int

The number of prefill tokens.

litert_lm_engine_settings_set_num_threads(nint, int)

Sets the number of threads for the CPU backend.

```
public static extern void litert_lm_engine_settings_set_num_threads(nint settings,
int numThreads)
```

Parameters

settings nint

The engine settings.

numThreads int

The number of threads.

litert_lm_engine_settings_set_parallel_file_section_loading(nint, bool)

Sets whether the engine should load different sections of the litertlm file in parallel. Defaults to [true](#).

```
public static extern void
litert_lm_engine_settings_set_parallel_file_section_loading(nint settings,
bool parallelFileSectionLoading)
```

Parameters

settings nint

The engine settings.

`parallelFileSectionLoading` bool

Whether to load in parallel.

`litert_lm_engine_settings_set_prefill_chunk_size(nint, int)`

Sets the prefill chunk size for the engine. Only applicable for CPU backend with dynamic models.

```
public static extern void litert_lm_engine_settings_set_prefill_chunk_size(nint settings, int prefillChunkSize)
```

Parameters

`settings` nint

The engine settings.

`prefillChunkSize` int

The prefill chunk size.

`litert_lm_engine_settings_set_supported_audio_lora_ranks(nint, int[], nuint)`

Sets the supported Audio LoRA ranks for the engine.

```
public static extern int  
litert_lm_engine_settings_set_supported_audio_lora_ranks(nint settings, int[]  
loraRanks, nuint numRanks)
```

Parameters

`settings` nint

The engine settings.

loraRanks int[]

An array of supported Audio LoRA ranks.

numRanks nuint

The number of ranks in the array.

Returns

int

0 on success, non-zero on failure.

litert_lm_engine_settings_set_supported_lora_ranks(nint, int[], nuint)

Sets the supported LoRA ranks for the engine.

```
public static extern int litert_lm_engine_settings_set_supported_lora_ranks(nint settings, int[] loraRanks, nuint numRanks)
```

Parameters

settings nint

The engine settings.

loraRanks int[]

An array of supported LoRA ranks.

numRanks nuint

The number of ranks in the array.

Returns

int

0 on success, non-zero on failure.

litert_lm_engine_settings_set_use_ringbuffers_local_attention(nint, bool)

Sets whether to use ringbuffers for local attention KV cache.

```
public static extern void  
litert_lm_engine_settings_set_use_ringbuffers_local_attention(nint settings, bool  
useRingbuffersLocalAttention)
```

Parameters

settings nint

The engine settings.

useRingbuffersLocalAttention bool

Whether to use ringbuffers for local attention.

Remarks

When enabled for supported models, a ringbuffer stores only necessary KV cache memory for local attention layers, minimizing memory usage. When disabled, memory is allocated for the full context length, enabling instant rewinding at the cost of higher memory usage.

Note: This feature is backend-agnostic in interface design, but currently only supported by the GPU Artisan backend. Enabling it on unsupported models or backends will be ignored with a warning.

Class NativeAPI.InputData

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.InputData
```

Inheritance

object ← NativeAPI.InputData

Methods

litert_lm_input_data_create(InputDataType, nint, nuint)

Creates a LiteRT LM Input Data. The caller is responsible for destroying the input data using [litert_lm_input_data_delete\(nint\)](#).

```
public static extern nint litert_lm_input_data_create(InputDataType type, nint data, nuint size)
```

Parameters

type [InputDataType](#)

The type of the input data.

data nint

The data pointer. The data is copied internally.

size nuint

The size of the data in bytes.

Returns

nint

A pointer to the created input data, or nint.Zero on failure.

litert_lm_input_data_delete(nint)

Destroys a LiteRT LM Input Data.

```
public static extern void litert_lm_input_data_delete(nint inputData)
```

Parameters

inputData nint

The input data to destroy.

Class NativeAPI.JsonResponse

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.JsonResponse
```

Inheritance

object ← NativeAPI.JsonResponse

Methods

litert_lm_json_response_delete(nint)

Destroys a LiteRT LM Json Response object.

```
public static extern void litert_lm_json_response_delete(nint response)
```

Parameters

response nint

The response to destroy.

litert_lm_json_response_get_string(nint)

Returns the JSON response string from a response object.

```
public static extern nint litert_lm_json_response_get_string(nint response)
```

Parameters

response nint

The response object.

Returns

nint

The response JSON string. The returned string is owned by the JsonResponse object and is valid only for its lifetime. Returns nint.Zero if **response** is nint.Zero.

Class NativeAPI.NoRepeatNgramConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.NoRepeatNgramConfig
```

Inheritance

object ← NativeAPI.NoRepeatNgramConfig

Methods

litert_lm_no_repeat_ngram_config_create()

Creates a LiteRT LM No Repeat Ngram Config with default values (`no_repeat_ngram_size` = 0, `window_size` = 0, which means no repeat ngram banning is disabled).

```
public static extern nint litert_lm_no_repeat_ngram_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

Remarks

When `no_repeat_ngram_size` is set greater than 0, any sequence of tokens (an ngram of that exact length) generated during decoding or present inside the window history can only occur at most once. If generating a candidate token would complete a repeating ngram, that candidate token's logit is set to -inf.

The caller is responsible for destroying the config using [litert_lm_no_repeat_ngram_config_delete\(nint\)](#).

litert_lm_no_repeat_ngram_config_delete(nint)

Destroys a LiteRT LM No Repeat Ngram Config.

```
public static extern void litert_lm_no_repeat_ngram_config_delete(nint config)
```

Parameters

config nint

The config to destroy.

litert_lm_no_repeat_ngram_config_set_no_repeat_ngram_size(nint, int)

Sets the no repeat ngram size for the no repeat ngram config.

```
public static extern void  
litert_lm_no_repeat_ngram_config_set_no_repeat_ngram_size(nint config,  
int noRepeatNgramSize)
```

Parameters

config nint

The config to modify.

noRepeatNgramSize int

The size of ngrams (consecutive token sequences) that are banned from repeating within the generation history window. If set > 0, when generating the next token would complete an already observed **no_repeat_ngram_size** sequence, the logit of the candidate token is set to -inf. If set <= 0, no repeat ngram banning is disabled. Negative values are automatically clamped to 0 during execution.

litert_lm_no_repeat_ngram_config_set_window_size(nint, int)

Sets the window size for the no repeat ngram config.

```
public static extern void litert_lm_no_repeat_ngram_config_set_window_size(nint  
config, int windowSize)
```

Parameters

`config` nint

The config to modify.

`windowSize` int

The maximum number of recent tokens in generation history to consider when checking for repeating ngrams. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be ≥ 0 ; negative values are clamped to 0. If `window_size` is greater than 0 but less than `no_repeat_ngram_size`, it is automatically clamped to `no_repeat_ngram_size` so that the ngrams can fit and be tracked.

Class NativeAPI.RepetitionPenaltyConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.RepetitionPenaltyConfig
```

Inheritance

object ← NativeAPI.RepetitionPenaltyConfig

Methods

litert_lm_repetition_penalty_config_create()

Creates a LiteRT LM Repetition Penalty Config with default values (`repetition_penalty` = 1.0f, `presence_penalty` = 0.0f, `frequency_penalty` = 0.0f, `window_size` = 0, which means all history with no penalties active).

```
public static extern nint litert_lm_repetition_penalty_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

Remarks

When multiple penalties are configured and active, the order of application to output logits during decoding is:

1. Multiplicative penalty (`repetition_penalty`)
2. Subtractive penalties (`presence_penalty` and `frequency_penalty`).

The caller is responsible for destroying the config using [litert_lm_repetition_penalty_config_delete\(nint\)](#).

litert_lm_repetition_penalty_config_delete(nint)

Destroys a LiteRT LM Repetition Penalty Config.

```
public static extern void litert_lm_repetition_penalty_config_delete(nint config)
```

Parameters

config nint

The config to destroy.

litert_lm_repetition_penalty_config_set_frequency_penalty(nint, float)

Sets the subtractive frequency penalty for the repetition penalty config.

```
public static extern void  
litert_lm_repetition_penalty_config_set_frequency_penalty(nint config,  
float frequencyPenalty)
```

Parameters

config nint

The config to modify.

frequencyPenalty float

A scalar subtracted from a token's logit, scaled linearly by the number of times that token has previously appeared inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

litert_lm_repetition_penalty_config_set_presence_penalty(nint, float)

Sets the subtractive presence penalty for the repetition penalty config.

```
public static extern void  
litert_lm_repetition_penalty_config_set_presence_penalty(nint config,
```

```
float presencePenalty)
```

Parameters

config nint

The config to modify.

presencePenalty float

A scalar subtracted from a token's logit if that token has appeared at least once inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

litert_lm_repetition_penalty_config_set_repetition_penalty(nint, float)

Sets the multiplicative repetition penalty for the repetition penalty config.

```
public static extern void  
litert_lm_repetition_penalty_config_set_repetition_penalty(nint config,  
float repetitionPenalty)
```

Parameters

config nint

The config to modify.

repetitionPenalty float

A multiplicative penalty applied to a token's logit if that token has appeared at least once inside the generated window history (e.g., 1.0 = no penalty, 1.2 = moderate penalty). Positive logits are divided by this parameter, and negative logits are multiplied (HuggingFace style). The parameter must be $\geq 1.0f$; values less than 1.0f are automatically clamped to 1.0f during execution.

litert_lm_repetition_penalty_config_set_window_size(nint, int)

Sets the window size for the repetition penalty config.

```
public static extern void litert_lm_repetition_penalty_config_set_window_size(nint  
config, int windowSize)
```

Parameters

config nint

The config to modify.

windowSize int

The maximum number of recent tokens in generation history to consider when computing penalization. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be ≥ 0 ; negative values are clamped to 0 during execution.

Class NativeAPI.Responses

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Responses
```

Inheritance

object ← NativeAPI.Responses

Methods

litert_lm_responses_delete(nint)

Destroys a LiteRT LM Responses object.

```
public static extern void litert_lm_responses_delete(nint responses)
```

Parameters

responses nint

The responses to destroy.

litert_lm_responses_get_num_candidates(nint)

Returns the number of response candidates.

```
public static extern int litert_lm_responses_get_num_candidates(nint responses)
```

Parameters

responses nint

The responses object.

Returns

int

The number of candidates.

litert_lm_responses_get_num_token_scores_at(nint, int)

Returns the number of tokens for which scores are present at a given index.

```
public static extern int litert_lm_responses_get_num_token_scores_at(nint responses,
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

int

The number of token scores. Returns 0 if index is out of bounds or no token scores are present.

litert_lm_responses_get_response_text_at(nint, int)

Returns the response text at a given index.

```
public static extern nint litert_lm_responses_get_response_text_at(nint responses,
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

nint

The response text. The returned string is owned by the Responses object and is valid only for its lifetime. Returns nint.Zero if index is out of bounds.

litert_lm_responses_get_score_at(nint, int)

Returns the score at a given index.

```
public static extern float litert_lm_responses_get_score_at(nint responses,
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

float

The score. Returns 0.0f if index is out of bounds or no score is present.

litert_lm_responses_get_token_length_at(nint, int)

Returns the token length at a given index.

```
public static extern int litert_lm_responses_get_token_length_at(nint responses,
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

int

The token length. Returns 0 if index is out of bounds or no token length is present.

litert_lm_responses_get_token_scores_at(nint, int)

Returns the token scores at a given index.

```
public static extern nint litert_lm_responses_get_token_scores_at(nint responses,
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

nint

A pointer to the internal array of token scores. Returns nint.Zero if index is out of bounds or no token scores are present.

litert_lm_responses_has_score_at(nint, int)

Returns whether the response contains a score at the given index.

```
public static extern bool litert_lm_responses_has_score_at(nint responses,  
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

bool

[true](#) if the score is available at the given index, [false](#) otherwise.

litert_lm_responses_has_token_length_at(nint, int)

Returns whether the response contains a token length at the given index.

```
public static extern bool litert_lm_responses_has_token_length_at(nint responses,  
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

bool

[true](#) if the token length is available at the given index, [false](#) otherwise.

litert_lm_responses_has_token_scores_at(nint, int)

Returns whether the response contains token scores at the given index.

```
public static extern bool litert_lm_responses_has_token_scores_at(nint responses,
int index)
```

Parameters

responses nint

The responses object.

index int

The index of the response.

Returns

bool

[true](#) if token scores are available at the given index, [false](#) otherwise.

Class NativeAPI.SamplerParams

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.SamplerParams
```

Inheritance

object ← NativeAPI.SamplerParams

Methods

litert_lm_sampler_params_create(SamplerType)

Creates LiteRT LM Sampler Parameters with a specific sampler type. The caller is responsible for destroying the parameters using [litert_lm_sampler_params_delete\(nint\)](#).

```
public static extern nint litert_lm_sampler_params_create(SamplerType type)
```

Parameters

type [SamplerType](#)

The sampler type to use.

Returns

nint

A pointer to the created parameters, or nint.Zero on failure.

litert_lm_sampler_params_delete(nint)

Destroys LiteRT LM Sampler Parameters.

```
public static extern void litert_lm_sampler_params_delete(nint @params)
```

Parameters

params nint

The parameters to destroy.

litert_lm_sampler_params_set_seed(nint, int)

Sets the seed.

```
public static extern void litert_lm_sampler_params_set_seed(nint @params, int seed)
```

Parameters

params nint

seed int

litert_lm_sampler_params_set_temperature(nint, float)

Sets the temperature.

```
public static extern void litert_lm_sampler_params_set_temperature(nint @params, float temperature)
```

Parameters

params nint

temperature float

litert_lm_sampler_params_set_top_k(nint, int)

Sets the top-k value.

```
public static extern void litert_lm_sampler_params_set_top_k(nint @params, int topK)
```

Parameters

params nint

topK int

litert_lm_sampler_params_set_top_p(nint, float)

Sets the top-p value.

```
public static extern void litert_lm_sampler_params_set_top_p(nint @params,  
float topP)
```

Parameters

params nint

topP float

Class NativeAPI.Session

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Session
```

Inheritance

object ← NativeAPI.Session

Methods

litert_lm_session_cancel_process(nint)

Cancels the current processing in the session.

```
public static extern void litert_lm_session_cancel_process(nint session)
```

Parameters

session nint

The session to cancel processing on.

litert_lm_session_delete(nint)

Destroys a LiteRT LM Session.

```
public static extern void litert_lm_session_delete(nint session)
```

Parameters

session nint

The session to destroy.

litert_lm_session_generate_content(nint, nint[], nuint)

Generates content from the input prompt.

```
public static extern nint litert_lm_session_generate_content(nint session, nint[] inputs, nuint numInputs)
```

Parameters

session nint

The session to use for generation.

inputs nint[]

An array of InputData structs representing the multimodal input.

numInputs nuint

The number of InputData structs in the array.

Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert_lm_responses_delete\(nint\)](#).

litert_lm_session_generate_content_stream(nint, nint[], nuint, nint, nint)

Generates content from the input prompt and streams the response via a callback. This is a non-blocking call that will invoke the callback from a background thread for each chunk.

```
public static extern int litert_lm_session_generate_content_stream(nint session, nint[] inputs, nuint numInputs, nint callback, nint callbackData)
```

Parameters

session nint

The session to use for generation.

inputs nint[]

An array of InputData structs representing the multimodal input.

numInputs nuint

The number of InputData structs in the array.

callback nint

The callback function ([NativeAPI.StreamCallback](#)) to receive response chunks.

callbackData nint

A pointer to user data that will be passed to the callback.

Returns

int

0 on success, non-zero on failure to start the stream.

litert_lm_session_get_benchmark_info(nint)

Retrieves the benchmark information from the session. The caller is responsible for destroying the benchmark info using [litert_lm_benchmark_info_delete\(nint\)](#).

```
public static extern nint litert_lm_session_get_benchmark_info(nint session)
```

Parameters

session nint

The session to get the benchmark info from.

Returns

nint

A pointer to the benchmark info, or nint.Zero on failure.

litert_lm_session_rewind_to_checkpoint(nint, string)

Rewinds the session to the given checkpoint label.

```
public static extern int litert_lm_session_rewind_to_checkpoint(nint session,
string label)
```

Parameters

session nint

The session to rewind.

label string

Label of the checkpoint to rewind to.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_rewind_to_step(nint, int)

Rewinds the session to a specific step number.

```
public static extern int litert_lm_session_rewind_to_step(nint session, int step)
```

Parameters

session nint

The session to rewind.

step int

The step number to rewind to.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_run_decode(nint)

Starts the decoding process for the model to predict the response based on the input prompt/query added after using [litert_lm_session_run_prefill\(nint, nint\[\], nuint\)](#). This is a blocking call and the function will return when the decoding process is done.

```
public static extern nint litert_lm_session_run_decode(nint session)
```

Parameters

session nint

The session to use.

Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert_lm_responses_delete\(nint\)](#).

litert_lm_session_run_decode_async(nint, nint, nint)

Starts the decoding process for the model to predict the response based on the input prompt/query added after using [litert_lm_session_run_prefill\(nint, nint\[\], nuint\)](#). This is a non-blocking call that will stream responses via a callback.

```
public static extern int litert_lm_session_run_decode_async(nint session, nint  
callback, nint callbackData)
```

Parameters

session nint

The session to use.

callback nint

The callback function ([NativeAPI.StreamCallback](#)) to receive response chunks.

callbackData nint

A pointer to user data that will be passed to the callback.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_run_prefill(nint, nint[], nuint)

Adds the input prompt/query to the model for starting the prefilling process. This is a blocking call and the function will return when the prefill process is done.

```
public static extern int litert_lm_session_run_prefill(nint session, nint[] inputs,
nuint numInputs)
```

Parameters

session nint

The session to use.

inputs nint[]

An array of InputData structs representing the multimodal input.

numInputs nuint

The number of InputData structs in the array.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_run_text_scoring(nint, string[], nuint, bool)

Scores the target text after the prefill process is done.

```
public static extern nint litert_lm_session_run_text_scoring(nint session, string[]
targetText, nuint numTargets, bool storeTokenLengths)
```

Parameters

session nint

The session to use.

targetText string[]

An array of target text strings to score.

numTargets nuint

The number of strings in the targetText array.

storeTokenLengths bool

Whether to store the token lengths of the target texts in the responses.

Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert_lm_responses_delete\(nint\)](#).

litert_lm_session_save_checkpoint(nint, string)

Saves the current state of the session to a checkpoint with the given label.

```
public static extern int litert_lm_session_save_checkpoint(nint session,
string label)
```

Parameters

`session` nint

The session to save checkpoint for.

`label` string

Label for the checkpoint.

Returns

int

0 on success, non-zero on failure.

Class NativeAPI.SessionConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.SessionConfig
```

Inheritance

object ← NativeAPI.SessionConfig

Methods

litert_lm_session_config_create()

Creates a LiteRT LM Session Config. The caller is responsible for destroying the config using [litert_lm_session_config_delete\(nint\)](#).

```
public static extern nint litert_lm_session_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_session_config_delete(nint)

Destroys a LiteRT LM Session Config.

```
public static extern void litert_lm_session_config_delete(nint config)
```

Parameters

config nint

The config to destroy.

litert_lm_session_config_set_apply_prompt_template(nint, bool)

Sets whether to apply prompt template for this session.

```
public static extern void litert_lm_session_config_set_apply_prompt_template(nint config, bool applyPromptTemplate)
```

Parameters

config nint

The config to modify.

applyPromptTemplate bool

Whether to apply prompt template.

litert_lm_session_config_set_audio_lora_path(nint, string)

Sets the path to the Audio LoRA weights file.

```
public static extern int litert_lm_session_config_set_audio_lora_path(nint config, string audioLoraPath)
```

Parameters

config nint

The config to modify.

audioLoraPath string

The path to the audio LoRA weights file.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_config_set_lora_path(nint, string)

Sets the path to the LoRA weights file.

```
public static extern int litert_lm_session_config_set_lora_path(nint config,
string loraPath)
```

Parameters

config nint

The config to modify.

loraPath string

The path to the text LoRA weights file.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_config_set_max_output_tokens(nint, int)

Sets the maximum number of output tokens per decode step for this session.

```
public static extern void litert_lm_session_config_set_max_output_tokens(nint
config, int maxOutputTokens)
```

Parameters

config nint

The config to modify.

maxOutputTokens int

The maximum number of output tokens.

litert_lm_session_config_set_sampler_params(nint, nint)

Sets the sampler parameters for this session config.

```
public static extern void litert_lm_session_config_set_sampler_params(nint config,  
nint samplerParams)
```

Parameters

config nint

The config to modify.

samplerParams nint

The sampler parameters to use.

Delegate NativeAPI.StreamCallback

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

Callback for streaming responses.

```
public delegate void NativeAPI.StreamCallback(nint callbackData, nint chunk)
```

Parameters

callbackData nint

A pointer to user-defined data passed to the stream function.

chunk nint

A pointer to the stream chunk object. It's only valid for the duration of the call.

Class NativeAPI.StreamChunk

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.StreamChunk
```

Inheritance

object ← NativeAPI.StreamChunk

Methods

litert_lm_stream_chunk_get_error(nint)

Gets the error message associated with this chunk, if any.

```
public static extern nint litert_lm_stream_chunk_get_error(nint chunk)
```

Parameters

chunk nint

Returns

nint

Remarks

Returns nint.Zero if there is no error.

litert_lm_stream_chunk_get_text(nint)

Gets the text content of the chunk.

```
public static extern nint litert_lm_stream_chunk_get_text(nint chunk)
```

Parameters

chunk nint

Returns

nint

Remarks

The returned string is owned by the chunk and is only valid as long as the chunk is valid. Returns nint.Zero if there is no text content in this chunk (e.g. if it is an error or metadata-only chunk).

litert_lm_stream_chunk_is_final(nint)

Returns [true](#) if this is the final chunk of the stream.

```
public static extern bool litert_lm_stream_chunk_is_final(nint chunk)
```

Parameters

chunk nint

Returns

bool

Class NativeAPI.SuppressTokensConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.SuppressTokensConfig
```

Inheritance

object ← NativeAPI.SuppressTokensConfig

Methods

litert_lm_suppress_tokens_config_create()

Creates a LiteRT LM Suppress Tokens Config with default values (an empty set of suppressed tokens, which means token suppression is disabled).

```
public static extern nint litert_lm_suppress_tokens_config_create()
```

Returns

nint

A pointer to the created config, or IntPtr.Zero on failure.

Remarks

When `suppress_tokens` is configured with one or more token IDs, the logits corresponding to those exact token IDs will be set directly to -inf during generation. This guarantees that those tokens can never be sampled by the model.

The caller is responsible for destroying the config using [litert_lm_suppress_tokens_config_delete\(nint\)](#).

litert_lm_suppress_tokens_config_delete(nint)

Destroys a LiteRT LM Suppress Tokens Config.

```
public static extern void litert_lm_suppress_tokens_config_delete(nint config)
```

Parameters

config nint

The config to destroy.

litert_lm_suppress_tokens_config_set_suppress_tokens(nint, int[]?, nuint)

Sets the list of token IDs to suppress for the suppress tokens config.

```
public static extern void litert_lm_suppress_tokens_config_set_suppress_tokens(nint config, int[]? suppressTokens, nuint numTokens)
```

Parameters

config nint

The config to modify.

suppressTokens int[]

An array of integer token IDs that should be banned from generation. During every decode step, each listed token ID's candidate logit will be forced to -inf. If **suppress_tokens** is [null](#) or **num_tokens** is 0, any previously set suppressed tokens are cleared and token suppression is disabled.

numTokens nuint

The number of token IDs in the **suppress_tokens** array.

Class NativeAPI.ThinkingConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.ThinkingConfig
```

Inheritance

object ← NativeAPI.ThinkingConfig

Methods

litert_lm_thinking_config_create()

Creates a default LiteRT LM Thinking Config (enabled with infinite budget -1). The caller is responsible for destroying the config using [litert_lm_thinking_config_delete\(nint\)](#).

```
public static extern nint litert_lm_thinking_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_thinking_config_delete(nint)

Destroys a LiteRT LM Thinking Config.

```
public static extern void litert_lm_thinking_config_delete(nint config)
```

Parameters

config nint

The config to destroy.

litert_lm_thinking_config_set_enable_thinking(nint, bool)

Sets whether thinking/reasoning generation is enabled.

```
public static extern void litert_lm_thinking_config_set_enable_thinking(nint config,
bool enableThinking)
```

Parameters

config nint

The config to modify.

enableThinking bool

Whether thinking is enabled.

litert_lm_thinking_config_set_thinking_token_budget(nint, int)

Sets the thinking token budget.

```
public static extern void litert_lm_thinking_config_set_thinking_token_budget(nint
config, int thinkingTokenBudget)
```

Parameters

config nint

The config to modify.

thinkingTokenBudget int

Budget for token-by-token reasoning generation (-1 for infinite).

Class NativeAPI.TokenUnion

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.TokenUnion
```

Inheritance

object ← NativeAPI.TokenUnion

Methods

litert_lm_token_union_delete(nint)

Destroys a LiteRT LM Token Union.

```
public static extern void litert_lm_token_union_delete(nint tokenUnion)
```

Parameters

tokenUnion nint

The token union to destroy.

litert_lm_token_union_get_ids(nint, out nint, out nuint)

Returns the token ids from a token union.

```
public static extern int litert_lm_token_union_get_ids(nint tokenUnion, out nint outTokens, out nuint outNumTokens)
```

Parameters

tokenUnion nint

The token union.

outTokens nint

A pointer to receive the internal array of token ids. The received pointer is valid only for the lifetime of the TokenUnion object.

`outNumTokens` `nuint`

A pointer to receive the number of token ids.

Returns

`int`

0 on success, non-zero if the type is not [Ids](#).

`litert_lm_token_union_get_string(nint)`

Returns the string value from a token union.

```
public static extern nint litert_lm_token_union_get_string(nint tokenUnion)
```

Parameters

`tokenUnion` `nint`

The token union.

Returns

`nint`

The string value, or `nint.Zero` if the type is not [String](#). The returned string is owned by the TokenUnion object and is valid only for its lifetime.

`litert_lm_token_union_get_type(nint)`

Returns the type of the token union.

```
public static extern TokenUnionType litert_lm_token_union_get_type(nint tokenUnion)
```

Parameters

`tokenUnion` nint

The token union.

Returns

[TokenUnionType](#)

The type of the token union.

Class NativeAPI.TokenUnions

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.TokenUnions
```

Inheritance

object ← NativeAPI.TokenUnions

Methods

litert_lm_token_unions_delete(nint)

Destroys a LiteRT LM Token Unions object.

```
public static extern void litert_lm_token_unions_delete(nint tokens)
```

Parameters

tokens nint

The token unions object to destroy.

litert_lm_token_unions_get_num_tokens(nint)

Returns the number of token unions in the collection.

```
public static extern nuint litert_lm_token_unions_get_num_tokens(nint tokens)
```

Parameters

tokens nint

The token unions object.

Returns

nuint

The number of token unions.

litert_lm_token_unions_get_token_at(nint, nuint)

Returns the token union at a given index from a collection.

```
public static extern nint litert_lm_token_unions_get_token_at(nint tokens,  
nuint index)
```

Parameters

tokens nint

The token unions collection.

index nuint

The index of the token union.

Returns

nint

A pointer to the token union at the given index, or nint.Zero if the index is out of bounds.
The caller is responsible for deleting the result using [litert_lm_token_union_delete\(nint\)](#).

Class NativeAPI.TokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.TokenizeResult
```

Inheritance

object ← NativeAPI.TokenizeResult

Methods

litert_lm_tokenize_result_delete(nint)

Destroys a LiteRT LM Tokenize Result.

```
public static extern void litert_lm_tokenize_result_delete(nint result)
```

Parameters

result nint

The tokenize result to destroy.

litert_lm_tokenize_result_get_num_tokens(nint)

Returns the number of token ids from a tokenize result.

```
public static extern nuint litert_lm_tokenize_result_get_num_tokens(nint result)
```

Parameters

result nint

The tokenize result.

Returns

nuint

The number of token ids.

litert_lm_tokenize_result_get_tokens(nint)

Returns the token ids from a tokenize result.

```
public static extern nint litert_lm_tokenize_result_get_tokens(nint result)
```

Parameters

result nint

The tokenize result.

Returns

nint

A pointer to the internal array of token ids. The returned pointer is valid only for the lifetime of the **result** object.